

Applied AI in Biomedicine: **Final Assignment**

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Outline



Background



Database



Assignment
objectives



Evaluation



Background: breast lesions

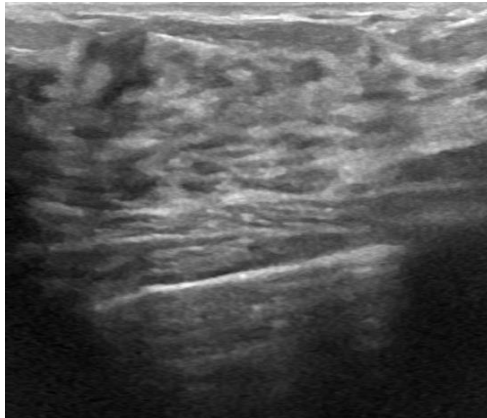
- Breast cancer is the most frequently diagnosed cancer among women worldwide, with more than 2.2 million new cases and over 650,000 deaths reported in 2020.
- Ultrasound, mammography, and magnetic resonance imaging are the primary imaging techniques used for breast evaluation.
- Ultrasound has become increasingly accessible and widely used.



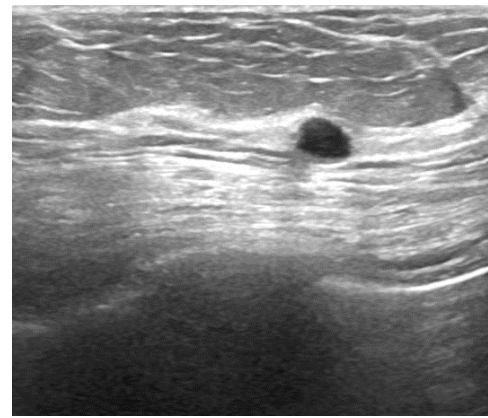
Background: breast lesions

Nevertheless, ultrasound remains a highly operator-dependent method, and the ability to detect key findings can vary depending on breast anatomy and tumor characteristics.

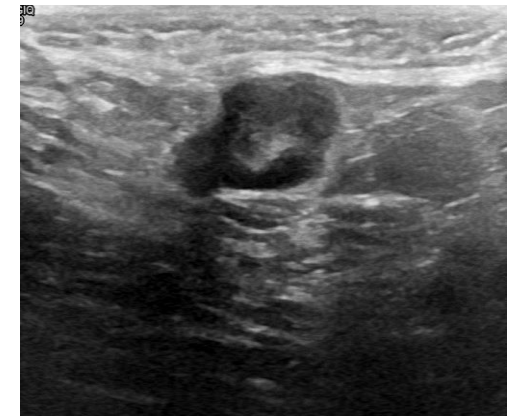
normal



benign



malignant





Database

1531 images from 968 patients:

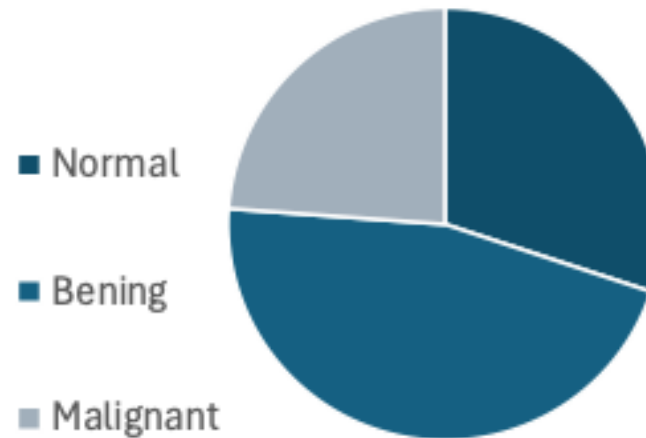
- 460 normal (N)
- 706 benign (B)
- 365 malignant (M)

Labels:

0 benign

1 malignant

2 normal





Database

If one patient has 3 lesions on the same image:

US	MASK	LABEL
933.png	933_mask.png	0
933.png	933_mask1.png	0
933.png	933_mask2.png	0

US	MASK	LABEL
933.png	933_mask.png	0

If one patient has many lesions on different images:

US	MASK	LABEL
1069_1.png	1069_1_mask.png	2
1069_2.png	1069_2_mask.png	2
1069_3.png	1069_3_mask.png	2
1069_4.png	1069_4_mask.png	0
1069_5.png	1069_5_mask.png	2
1069_6.png	1069_6_mask.png	0
1069_7.png	1069_7_mask.png	0
1069_8.png	1069_8_mask.png	2
1069_9.png	1069_9_mask.png	2
1069_10.png	1069_10_mask.png	0
1069_11.png	1069_11_mask.png	2
1069_12.png	1069_12_mask.png	2
1069_13.png	1069_13_mask.png	0
1069_14.png	1069_14_mask.png	2



Database

Example:

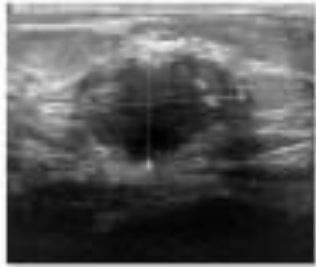
159.png



159_mask.png



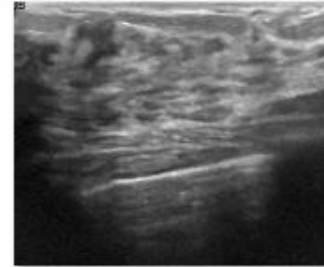
160.png



160_mask.png



1.png



1_mask.png



training_metadata.xlsx

	A	B	C
1	US	MASK	LABEL
147	159.png	159_mask.png	1
148	160.png	160_mask.png	1



Assignment objectives

1. Classification & segmentation

1a. Classify the lesions into the 3 classes (N/B/M)

[→ 3-class classification model]

1b. Segment the lesions for B and M classes

[→ 2-class segmentation model]

→ Evaluate both task performance with the appropriate metrics (for the classification task, show also the 3x3 confusion matrix)



Assignment objectives

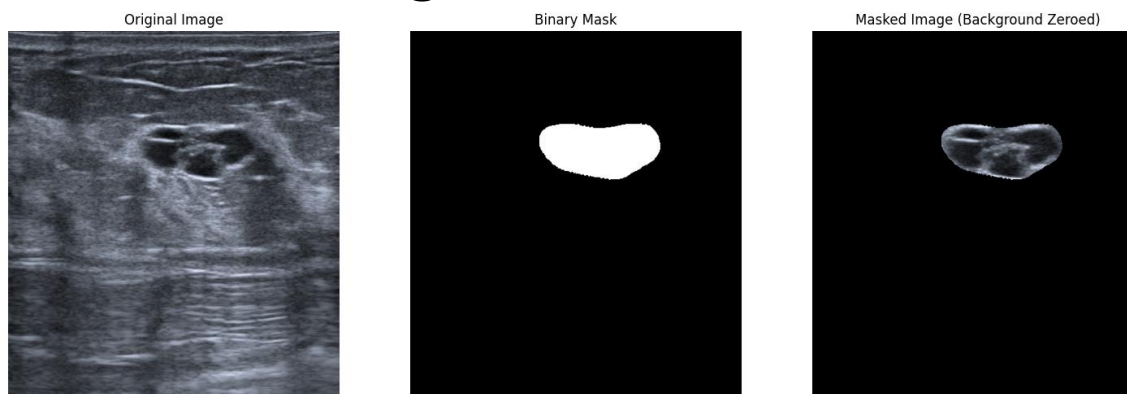
2. Segmentation & classification

2a. Segment all lesions (N/B/M)

[→ 3-class segmentation model]

2b. Classify the lesions B vs. M using the masked image

[→ 2-class classification model]



→ Evaluate the performance both task performance with the appropriate metrics (for the classification task, show also the 3x3 confusion matrix)



Evaluation

- Report
- Scoring
- Deadlines
- Exam dates



Evaluation - Report

Write a report in paper format:

- introduction (brief but with references),
- materials and methods,
- results,
- discussion and conclusion
- ~6 pages double column format (not strict length!)



Evaluation - Report

In the report

- Describe in detail the pre-processing
- Describe the models chosen and their training
- Show in the appendix images with loss and accuracy (for classification) or loss and Dice score (for segmentation) during epochs
- If you use data augmentation, show in the appendix examples of augmented images
- Show examples of segmentation results (original image, true mask and predicted mask)
- Justify your choices in a sensitive way and provide a clear interpretation of the results obtained



Evaluation - Report

Describe and explain:

- The four developed classification and segmentation models
- Compare the two approaches (classification first and then segmentation, segmentation first and then classification) from a methodological point of view and in terms of obtained results

PS: write in the report all you want to be evaluated
What is not in the report will not be considered
(even if in the code)



Evaluation - Scoring

- Submit report + code (training code and the final working model) one week before the exam.
 - Once the report and code have been submitted, you will be provided with a test set without labels: submit the labels created by your model within the next 24h in the same format as the labels provided to you in the training set
- You will be provided with a test set scoring (before or during the oral exam), if the balanced accuracy of the test set is > 0.75 → extra points**
- In the exam you will have to justify the choices made and to interpret the results obtained. You will also have to answer questions regarding the material seen during the lectures and practical sessions.



Evaluation - Deadlines

Possible exam days:

- January 8
- January 9
- January 15
- January 16
- January 22
- January 23
-

PS: assignment to be done in groups (3 students)
Oral exam together, book one slot per group

8	Report + code
9	Test set predictions
10	
11	
12	
13	
14	
15	exam

Test set (without labels)

Scoring Test set



Evaluation - Exam dates

Doodle:

I will prepare a doodle and a Google sheet:

- The doodle will have 75-minute slots
- Book one slot per group (choose a group name)
- Write the group name plus the participant names in the Google sheet



Data download

You can access data at the following link:

https://drive.google.com/drive/folders/1EkuEi0h2G_nVuD08Z3dUJpXl63sgY79?usp=drive_link

In the folder you'll find:

"training_images" folder → contains images and ground-truth masks

"training_metadata.xlsx" file → contains filenames of images and masks + label